

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Seventh Semester of B. Tech. (EE) Examination****December 2015****EE405 Power System Protection****Date: 16/12/2015, Wednesday****Time: 10.00 a.m. to 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q – 1** Choose the correct option**[07]**

- I. The distribution system shown in the figure 1 is to be protected by overcurrent system of protection. For proper fault discrimination, directional overcurrent relays will be required at locations

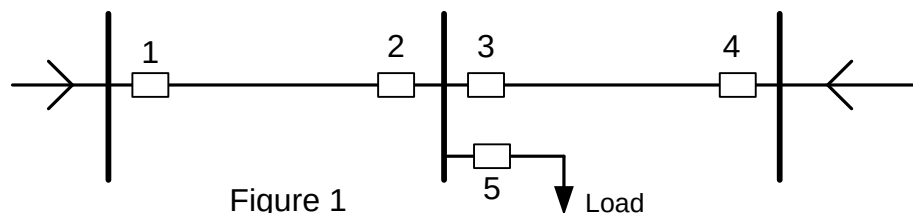


Figure 1

- (a) 1 and 4 (b) 2 and 3 (c) 1, 4 and 5 (d) 2, 3 and 5
- II. Consider the diagram drawn in figure 2

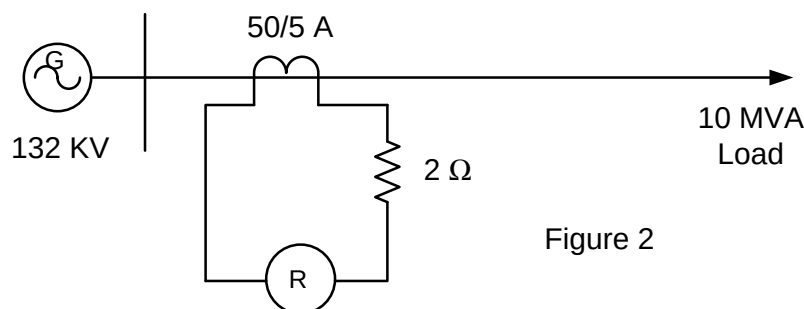


Figure 2

If the relay is rated at 5 VA, which of the following is the correct VA for CT.

- (a) 34 (b) 40 (c) 45 (d) 37
- III. The most efficient torque-producing actuating structure for induction-type relays is
- (a) shaded pole structure. (b) watt-hour meter structure.
(c) induction-cup structure. (d) single induction loop structure.
- IV. Differential relays are used for protection of equipment against
- (a) internal faults. (b) overcurrent.
(c) reverse current. (d) reverse power.

- V. In distance protection, the relay measures
- +ve sequence impedance of the line from relay up to the fault point.
 - ve sequence impedance of the line from relay up to the fault point.
 - self impedance of the line from relay up to the fault point.
 - zero sequence impedance of the line from relay up to the fault point.
- VI. Both voltage and current signals are required for
- a plain overcurrent relay.
 - a differential relay.
 - a directional relay.
 - a biased differential relay.
- VII. In carrier current protection the purpose of the wave-trap is for
- trapping power frequency waves.
 - trapping high frequency waves entering into generators/transformer unit.
 - both (a) and (b).
 - none of the above.

Q – 2. A Figure 3 shows a portion of a power system wherein IDMT over current relays giving standard IDMT characteristic are used. If the TMS of R_4 and R_5 are set at 0.1 and PS of R_5 is set at 175%, determine the plug setting of relays R_1 to R_4 and time setting of relays R_1 to R_3 . [08]

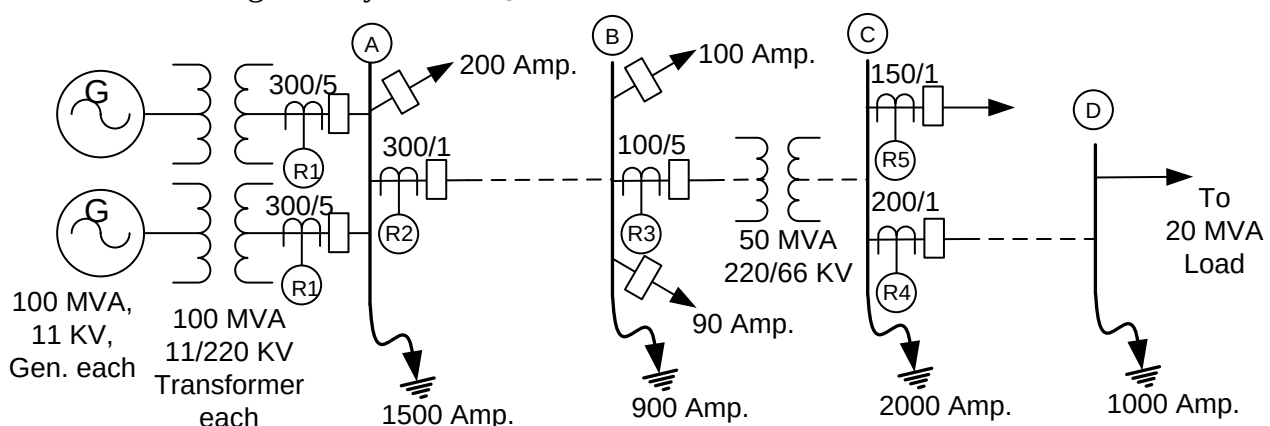


Figure 3

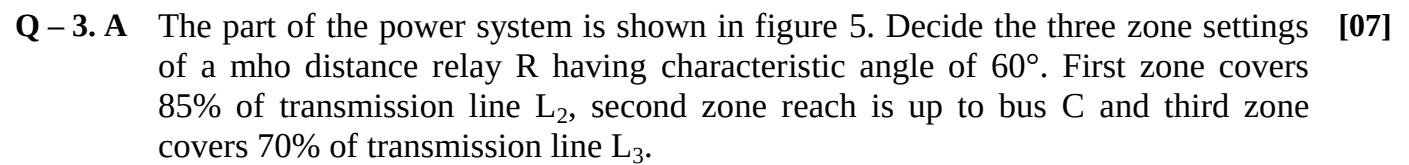
B Give Reason

- Directional over current relays are used at load end of parallel feeders.
- Dead zone is present in directional over current relay.

OR

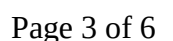
Q – 2. A Give Reason

- Ground over current relays can be made more sensitive than phase over current relays. However, their sensitivity is limited if they are connected in the residual circuit of three line CTs.
- Inverse definite minimum time lag over current relays gives better discrimination than inverse-time overcurrent relays.



(i) 132 KV Lines L_1 and L_2
Impedance: $0.02 + j 0.06$ ohm / km / ph. Line length: 150 Km.

(ii) 66 KV line L_3 :
Impedance: $0.04 + j 0.08$ ohm / Km / ph. Line length: 100 Km



- B** With neat diagram explain the blocking carrier scheme used with carrier aided distance protection. [07]

OR

- Q – 3. A** Derive the expression for estimation of phasors using a full-cycle Fourier algorithm. [08]

- B** Explain dynamic testing of relays. [06]

SECTION – II

- Q – 4. A** Choose the correct option [03]

- I. Which of the following is true regarding relay
- Mho relay is inherently directional.
 - Instantaneous O/C relay is used for protection of long O/H lines.
 - Negative sequence relay is used for protection of transformer.
 - Carrier current is used for protection of 11KV feeders.
- II. Consider the generator stator winding protected as per figure 6.

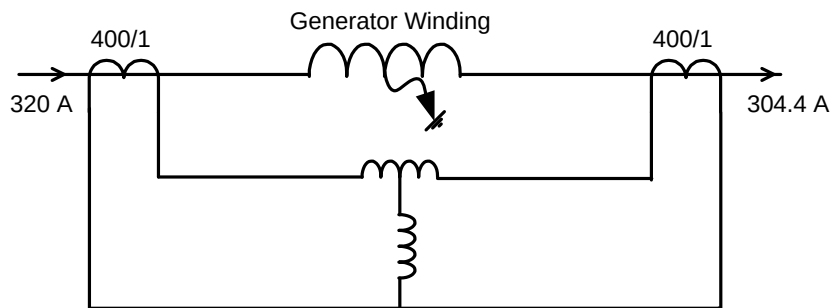


Figure 6

The ratio of operating current to restraining current for above drawn condition is...

- (a) 3% (b) 5% (c) 8% (d) 11%
- III. For fault F, as shown in figure 7, the impedance seen by distance relay R is...

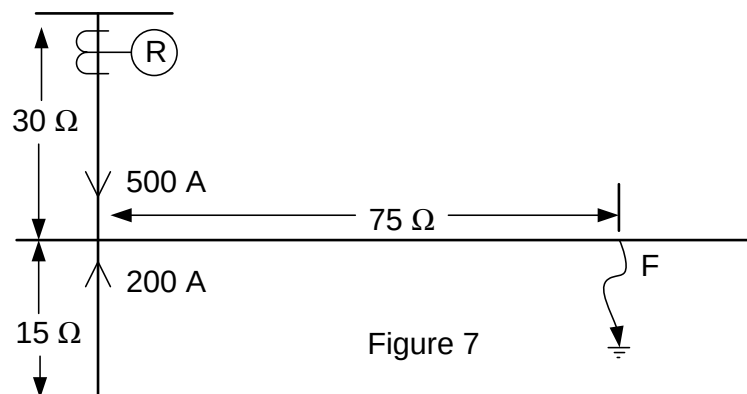


Figure 7

- (a) 160 Ω (b) 150 Ω (c) 135 Ω (d) 120 Ω

B Write the consequence of following statement. [04]

- I. In class A protection, if field breaker is not opened immediately.
- II. If there is an open circuit in the field circuit of generator.
- III. If hot characteristic curve overlap with starting characteristic of induction motor.
- IV. In differential protection, if polarity of CTs are wrongly connected.

Q – 5. A A 15.75 KV, 3 phase, 50 Hz, 210 MVA, star connected generator is protected by the simple differential protection. The CTs used are 7500/5 A. The relay is set to operate for a current of 0.15 A. [07]

The p.u. reactance of generator is 8%. Under through fault condition, the CT₂ secondary voltage is 75% of CT₁ secondary voltage.

The relay having a resistive impedance of 100 ohms is connected to 100 meters from CT₁. The lead wire has a resistance of 0.5 ohm per 100 meters. The distance between the two sets of CTs is 250 meters.

Determine the stabilizing resistance required to be connected in series with relay to have a stability factor of 3 for this fault condition.

B I. Discuss the effect of fault resistance in the performance of distance relay. [03]
How it is overcome by proper choice of relay?

II. With relevant characteristic and control circuit explain how the distance relay operation is blocked during power swing. [04]

OR

Q – 5. A Following are the details of 3 phase, 1000 HP, 3.3 KV Induction motor. [07]

- i. Rated p.f.:0.85
- ii. Efficiency: 85%
- iii. Starting time: 6 seconds
- iv. Starting Current: 5.5 times the rated current
- v. CT ratio: 200/1 Amp.
- vi. Motor can able to take 110% of rated current continuously.

Relay: Rating =1 amp.

- a) Thermal O/L Unit: 70%-130% of 1 Amp in step of 10%
- b) Instantaneous O/C Unit: 400%-2000% of 1 Amp in step of 50%
- c) NPS Unit: 100% -600% of 1 Amp in steps of 50%
- d) Stalling Unit:

Current Setting Range: 100-600% of 1 Amp in step of 30%

Time Setting Range: 4-30 seconds.

(Safe stalling time of I.M. Hot=13 sec; Cold=21 Sec.)

- e) Under Voltage Relay: Range 70-100% of 110 V. (P.T. 3.3 KV/110V)

Suggest the setting of above mentioned relay for proper protection of I.M.

B A transformer is protected by biased differential relay. The relevant details are as per following. [07]

i. Transformer:

500 MVA, 400/132 KV, 3 phase, 50 Hz, DY₁₁, Neutral solidly grounded.

Maximum through fault current is 13 times the rated current.

$I_{\text{mag Inrush}} = 10X I_{\text{Rated}}$, $I_0 = 4\%$ of I_{Rated} .

On load Tap changer = -10% to +7.5% in step of 2.5% ,

ii. Current transformers:

L.V. Side: 2500/1A Y-connected, H.V. Side: 750/1 A Y-connected.

Interposing CTs connected on L.V. side. CTs are of class 3

iii. Biased Differential Relay: (Relay rating 1 Amp.)

Basic setting: 5-25% of 1 amp in step of 5%

Bias setting: 10-50% in step of 10%

H.S. inst. setting 800 to 1600% of 1 Amp in step of 100%

Find out the ICT ratio and draw the detailed 3-Φ diagram of protective scheme. Also suggest the relay setting.

Q – 6. A Draw neat and well labeled diagram for duplicate busbar protection scheme. In this diagram show operation of relay during any in-zone fault. [07]

B I. Give reason: A stabilizing resistance is connected in series with the relay coil in differential protection of generator. [02]

II. Why stator 100% earth-fault protection is required for large generators? With neat diagrams, explain the stator 100% earth-fault protection. [05]

OR

Q – 6. A I. What is meant by burden of a CT? [02]

II. Give reason: knee point voltage of protective CT must be high, where as knee point voltage of measuring CT must be low. [03]

III. Explain the term 5P15 in connection with CT. [02]

B I. Explain why a delta-star power transformer must have the protective CTs connected in star-delta for differential relaying. [04]

II. Earth-fault relays employed for protection of the delta winding of a transformer are of instantaneous type while those on the star side have to be time graded. [03]
